


# 3 Fundamental Data Structures

*Efficient Algorithms · Wintersemester 2026–27*

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CS566 (Wintersemester 2026–27)
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# Learning Outcomes

**Unit 3:** ***Fundamental Data Structures***

1. Understand and demonstrate the difference between *abstract data type (ADT)* and its *implementation*
2. Be able to define the ADTs *stack*, *queue*, *priority queue* and *dictionary / symbol table*
3. Understand *array*-based implementations of stack and queue
4. Understand *linked lists* and the corresponding implementations of stack and queue
5. Know *binary heaps* and their performance characteristics
6. Understand *binary search trees* and their performance characteristics
7. Know high-level idea of basic *hashing strategies* and their performance characteristics

# Outline

## 3 Fundamental Data Structures


3.1 Stacks & Queues
3.2 Resizable Arrays
3.3 Priority Queues & Binary Heaps
3.4 Operations on Binary Heaps
3.5 Symbol Tables
3.6 Binary Search Trees
3.7 Ordered Symbol Tables
3.8 Balanced BSTs
3.9 Hashing – Basics
3.10 Universal Hashing

# Clicker Question



What's the running time (on our word-RAM model with word size $w$) of this Java instruction?

```
Object[] A = new Object[n];
```

| | | | | | |
|---|---|---|---|---|---|
| **A** | $1$ | **D** | $\Theta(w)$ | **G** | $\Theta(n \log n)$ |
| **B** | $\Theta(1)$ | **E** | $\Theta(n/w)$ | **H** | $\Theta(nw)$ |
| **C** | $\Theta(\log n)$ | **F** | $\Theta(n)$ | **I** | $\Theta(n^2)$ |



→ *sli.do/effalg*

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| **C** | ~~$\Theta(\log n)$~~ | **F** | $\Theta(n)$ ✓ | **I** | ~~$\Theta(n^2)$~~ |



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# Recap: The Random Access Machine

- Data structures make heavy use of pointers and dynamically allocated memory.

- Recall: Our RAM model supports
  - basic pseudocode ($\approx$ simple Python/Java code)
  - creating arrays of a fixed/known size.
  - creating instances (objects) of a known class.

# Recap: The Random Access Machine

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  - creating arrays of a fixed/known size.
  - creating instances (objects) of a known class.



Python abstracts this away!

no predefined capacity!

**There are *no* *arrays* in Python, only its built-in *lists*.**

But: Python *implementations create* lists based on fixed-size arrays (stay tuned!)



Python ≠ RAM: Not every built-in Python instruction runs in $O(1)$ time!

# 3.1 Stacks & Queues

# Abstract Data Types

**abstract data type (ADT)**

- list of supported operations
- **what** should happen
- **not:** how to do it
- **not:** how to store data

≈ Java `interface`, Python `ABC`s (with comments)

abstract base classes

VS.

**data structures**

- specify exactly **how** data is represented
- **algorithms** for operations
- has concrete costs (space and running time)

≈ Java/Python `class` (non abstract)

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*Why separate?*

- Can swap out implementations ⇝ "drop-in replacements"

⇝ **reusable code**!

- (Often) better abstractions
- Prove generic lower bounds ( ⇝ Unit 4)

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# Clicker Question



Which of the following are examples of abstract data types?

- **A** ADT
- **B** Stack
- **C** Deque
- **D** Linked list
- **E** binary search tree
- **F** Queue
- **G** resizable array
- **H** heap
- **I** priority queue
- **J** dictionary/symbol table
- **K** hash table



→ *sli.do/effalg*

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Which of the following are examples of abstract data types?



- **A** ~~ADT~~
- **B** Stack ✓
- **C** Deque ✓ dequeue
- **D** ~~Linked list~~
- **E** ~~binary search tree~~
- **F** Queue ✓
- **G** ~~resizable array~~
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→ *sli.do/effalg*

# Stacks



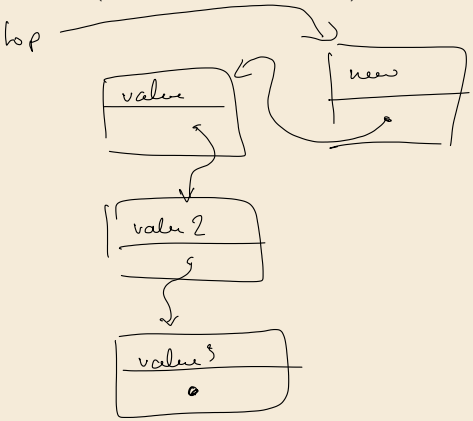
**Stack ADT**

- `top()`
  Return the topmost item on the stack
  Does not modify the stack.
- `push(`$x$`)`
  Add $x$ onto the top of the stack.
- `pop()`
  Remove the topmost item from the stack
  (and return it).
- `isEmpty()`
  Returns `true` iff stack is empty.
- `create()`
  Create and return an new empty stack.

# Linked-list implementation for Stack

**Invariants:**

- maintain pointer *top* to topmost element
- each element points to the element below it (or `null` if bottommost)




```
1  class Node
2      value
3      next
4
5  class Stack
6      top := null
7      procedure top():
8          return top.value
9      procedure push(x):
10         top := new Node(x, top)
11     procedure pop():
12         t := top()
13         top := top.next
14         return t
```

# Linked-list implementation for Stack – Discussion

**Linked stacks:**

- require $\Theta(n)$ space when $n$ elements on stack
- All operations take $O(1)$ time

- require $\Theta(n)$ space when $n$ elements on stack

Can we avoid extra space for pointers?

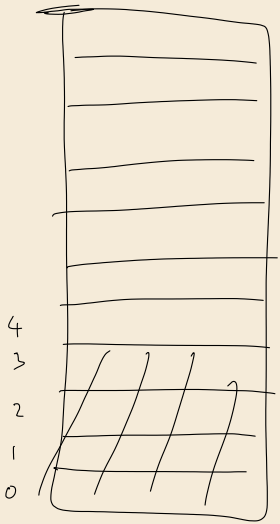
# Array-based implementation for Stack

If we want no pointers $\rightsquigarrow$ array-based implementation

**Invariants:**

- maintain array $S$ of elements, from bottommost to topmost
- maintain index *top* of position of topmost element in S.

top = 3



# Array-based implementation for Stack

If we want no pointers $\rightsquigarrow$ array-based implementation

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What to do if stack is full upon `push`?

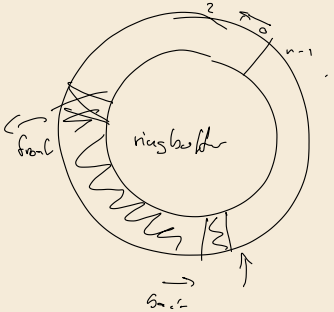
**Array stacks:**

- require *fixed capacity* $C$ (decided at creation time)!
- require $\Theta(C)$ space for a capacity of $C$ elements
- all operations take $O(1)$ time

# Queues

**Operations:**

- `enqueue(`$x$`)`
  Add $x$ at the end of the queue.
- `dequeue()`
  Remove item at the front of the queue and return it.






Implementations similar to stacks.

# Bags

*What do Stack and Queue have in common?*

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*What do Stack and Queue have in common?*



They are special cases of a ***Bag***!

**Update Operations:**

- `insert(`$x$`)`
  Add $x$ to the items in the bag.
- `delAny()`
  Remove any one item from the bag and return it.
  (Not specified which; any choice is fine.)
- roughly similar to Java's `java.util.Collection`
  Python's `collections.abc.Collection`
- always support iterating over content (read only)

Sometimes it is useful to *state* that order is irrelevant ⇝ Bag
Implementation of Bag usually just a Stack or a Queue

# 3.2 Resizable Arrays

# *Digression* – Arrays as ADT ?

Arrays can also be seen as an ADT!

**Array operations:**

- `create(`$n$`)` *Java*: `A = new int[`$n$`];` *Python:* `A = [0] *` $n$
  Create a new array with $n$ cells, with positions $0, 1, \ldots, n-1$;
  we write $A[0..n) = A[0..n-1]$
- `get(`$i$`)` *Java/Python*: `A[`$i$`]`
  Return the content of cell $i$
- `set(`$i$`,`$x$`)` *Java/Python*: `A[`$i$`] =` $x$`;`
  Set the content of cell $i$ to $x$.

⇝ Arrays have *fixed* size (supplied at creation). (≠ `lists` in Python)

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Arrays can also be seen as an ADT! . . . but are commonly seen as specific data structure

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⇝ Arrays have *fixed* size (supplied at creation). (≠ `lists` in Python)

Usually directly implemented by compiler + operating system / virtual machine.



**Difference to "real" ADTs:** ***Implementation usually fixed***
to "a contiguous chunk of memory".

# Doubling trick

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implement stack on arrays w/o capacity limit

**Invariants:**

- maintain array $S$ of elements, from bottommost to topmost
- maintain index *top* of position of topmost element in $S$
- maintain capacity $C = S.length$ so that $\frac{1}{4}C \leq n \leq C$

⇝ can always push more elements!

# Doubling trick

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$\rightsquigarrow$ can always push more elements!

*How to maintain the last invariant?*

- before `push`
  If $n = C$, allocate new array of size $2n$, copy all elements.
- after `pop`
  If $n < \frac{1}{4}C$, allocate new array of size $2n$, copy all elements.

$\rightsquigarrow$ ***"Resizing Arrays"***

an implementation technique, not an ADT!

# Clicker Question



Which of the following statements about resizable array that currently stores $n$ elements is correct?

**A** The elements are stored in an array of size $2n$.

**B** Adding or deleting an element at the end takes constant time.

**C** A sequence of $m$ insertions or deletions at the end of the array takes time $O(n + m)$.

**D** Inserting and deleting any element takes $O(1)$ amortized time.



→ *sli.do/effalg*

# Amortized Analysis

- Any individual operation `push` / `pop` can be expensive!
  $\Theta(n)$ time to copy all elements to new array.
- **But:** One expensive operation of cost $T$ means $\Omega(T)$ next operations are cheap!




Notation:

$c_i$ = actual cost of operation $i$

$\Phi_i$ = "potential" after operation $i$

$a_i$ = amortized cost of operation $i$

$:= c_i + \alpha(\Phi_i - \Phi_{i-1})$

scaling constant

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Goal: Prove that $a_i \leq A$

$$\sum_{i=1}^{m} a_i = \sum_{i=1}^{m} \left(c_i - \alpha(\Phi_i - \Phi_{i-1})\right) = \sum_{i=1}^{m} c_i + \alpha(\Phi_m - \Phi_0)$$

$\leq m \cdot A$

$$\Rightarrow \sum_{i=1}^{m} c_i$$

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distance to boundary — since $n \le C \le 4n$

**Formally:** consider "credits/potential" $\Phi = \min\{n - \frac{1}{4}C, C - n\} \in [0, 0.6n]$

- amortized cost of an operation = actual cost (array accesses) $- 4 \cdot$ change in $\Phi$
  - cheap `push`/`pop`: actual cost 1 array access, consumes $\le 1$ credits $\rightsquigarrow$ amortized cost $\le 5$
  - copying `push`: actual cost $2n + 1$ array accesses, creates $\frac{1}{2}n + 1$ credits $\rightsquigarrow$ amortized cost $\le 5$
  - copying `pop`: actual cost $2n + 1$ array accesses, creates $\frac{1}{2}n - 1$ credits $\rightsquigarrow$ amortized cost 5

$\rightsquigarrow$ **sequence** of $m$ operations: total actual cost $\le$ total amortized cost + final credits

here: $\le \quad 5m \quad + \quad 4 \cdot 0.6n \quad = \Theta(m + n)$

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# Deamortized Resizable Arrays

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# Deamortized Resizable Arrays

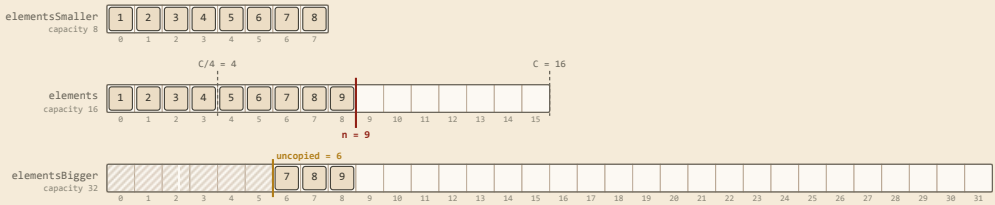
*What if we need $O(1)$ **worst case** time?*

- It's possible to *de-amortize* the resizing arrays solution!
- maintain **3 arrays**: $S$ (as before) and $S_2$ and $S_{1/2}$
  of twice and half the size of $S$

# Deamortized Resizable Arrays

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- maintain **3 arrays**: $S$ (as before) and $S_2$ and $S_{1/2}$
  of twice and half the size of $S$




- write operations go to all 3 arrays
- upon resize, "shift" arrays up/down $\rightsquigarrow$ $S_2$ resp. $S_{1/2}$ become new $S$
  - allocate new array, but **delay filling it with elements** ← general strategy!
  - every insert or delete *copies 2 slots* from last resize

$\rightsquigarrow$ by time for next resize, we have caught up and $S_2$ resp. $S_{1/2}$ ready to use

# Deamortized Resizable Arrays – Analysis

**Analysis:**

assuming memory allocation in $O(1)$ $\rightsquigarrow$ needs to be uninitialized!

- $O(1)$ worst case time for read/write by index, push, and pop!
- up to 7 array accesses per operation
- up to $14n$ space

other time-space trade-offs possible

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other time-space trade-offs possible

⚠ we assume here that array allocation is $O(1)$ time (no initialization!)

# *Rabbit Hole:* Can we do this more space-efficiently?

- It might appear as if every efficient implementation of a stack needs $\Omega(n)$ extra space on top of space for storing the $n$ elements in the stack.

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# *Rabbit Hole:* Can we do this more space-efficiently?

- It might appear as if every efficient implementation of a stack needs $\Omega(n)$ extra space on top of space for storing the $n$ elements in the stack.
- But this is not true!
- Can get operations in $O(1)$ worst-case time with $O(\sqrt{n})$ extra space at any time (!)
  - Maintain a collection of small arrays (plus header with pointers to them)
  - Clever choice of block sizes guarantees
    $O(\sqrt{n})$ blocks of $O(\sqrt{n})$ elements throughout
    and fast calculation of address for an index.

    imaginary "superblocks" of sizes $2^k$, $k = 0, 1, \ldots, \lg n$
    $k$th superblock consists of $2^{k/2}$ actual blocks of $2^{k/2}$ elements each.
  - $O(\sqrt{n})$ extra space is best possible



**Brodnik, Carlsson, Demaine, Munro, Sedgewick**: *Resizable Arrays in Optimal Time and Space,* WADS **1999**

# 3.3 Priority Queues & Binary Heaps

# Clicker Question



What is a heap-ordered tree?

**A** A tree in which every node has exactly 2 children.

**B** A tree where all keys in the left subtree are smaller than the key at the root and all keys in the right subtree are bigger than the key at the root.

**C** A tree where all keys in the left subtree and right subtree are smaller than the key at the root.

**D** An tree that is stored in the heap-area of the memory.



→ *sli.do/effalg*

# Priority Queue ADT

Now: elements in the bag have different *priorities*.

**(Max-oriented) Priority Queue (MaxPQ):**

- `construct(`$A$`)`
  Construct from from elements in array $A$.
- `insert(`$x$`,`$p$`)`
  Insert item $x$ with priority $p$ into PQ.
- `max()`
  Return item with largest priority. (Does not modify the PQ.)
- `delMax()`
  Remove the item with largest priority and return it.
- `changeKey(`$x$`,`$p'$`)`
  Update $x$'s priority to $p'$.
  Sometimes restricted to *increasing* priority.
- `isEmpty()`

Fundamental building block in many applications.



# Priority Queue ADT – min-oriented version

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- `~~max~~()`
  Return item with ~~largest~~ priority. (Does not modify the PQ.)
- `del~~Max~~()`
  Remove the item with ~~largest~~ priority and return it.
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  Update $x$'s priority to $p'$
  Sometimes restricted to ~~in~~*creasing* priority.
- `isEmpty()`

Fundamental building block in many applications.



# PQ implementations

**Elementary implementations**

- unordered list $\rightsquigarrow$ $\Theta(1)$ `insert`, but $\Theta(n)$ `delMax`
- sorted list $\rightsquigarrow$ $\Theta(1)$ `delMax`, but $\Theta(n)$ `insert`

# PQ implementations

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*Can we get something between these extremes? Like a "slightly sorted" list?*

# PQ implementations

**Elementary implementations**

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- sorted list $\rightsquigarrow$ $\Theta(1)$ `delMax`, but $\Theta(n)$ `insert`

*Can we get something between these extremes?* *Like a "slightly sorted" list?*

**Yes!** *Binary heaps.*

***Array view***

Heap = array $A[1..n]$ with
$\forall i \in [2..n] : A[\lfloor i/2 \rfloor] \geq A[i]$

$\Rightarrow$ max in $i = 1$

store nodes
in level order
in $A[1..n]$

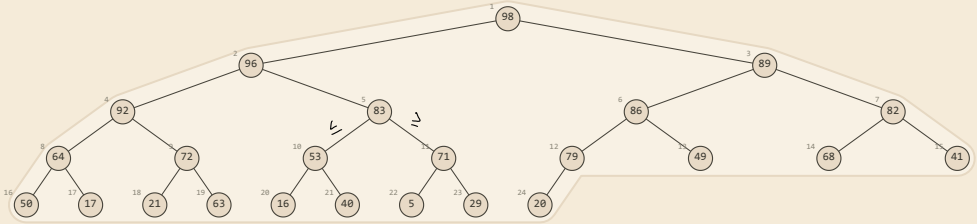
***Tree view***

Heap = tree that is
*(i)* a complete binary tree
*(ii)* heap ordered

all but last level full
last level flush left

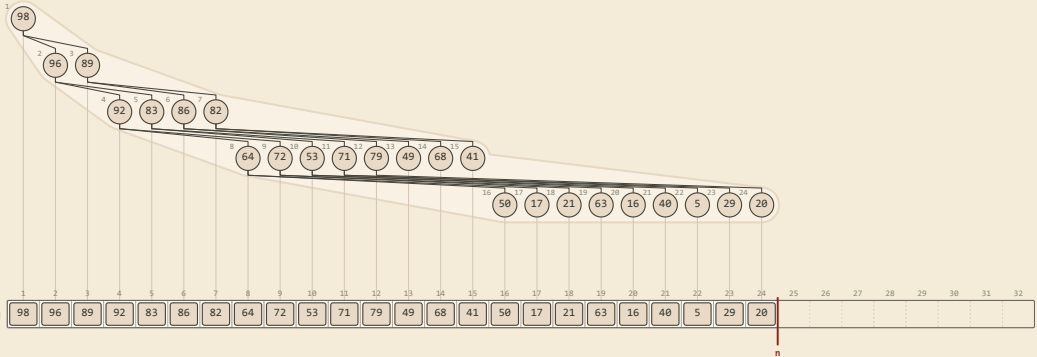
parent $\geq$ children

# Binary heap example

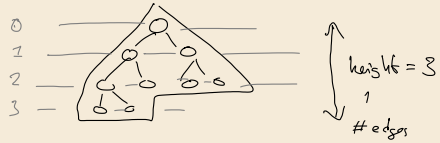





# Binary heap example



# Why heap-shaped trees?




**Why complete binary tree shape?**

- only one possible tree shape ⇝ keep it simple!
- complete binary trees have minimal height among all binary trees
- simple formulas for moving from a node to parent or children:

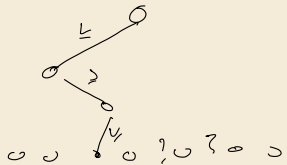
  For a node at index $k$ in $A$ ⚠ Recall: nodes at indices $[1..n]$
  - parent at $\lfloor k/2 \rfloor$ (for $k \geq 2$)
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  - right child at $2k + 1$



**Why heap ordered?**

- Maximum must be at root! ⇝ `max()` is trivial!
- But: Sorted only along paths of the tree; leaves lots of leeway for fast inserts

how? . . . stay tuned

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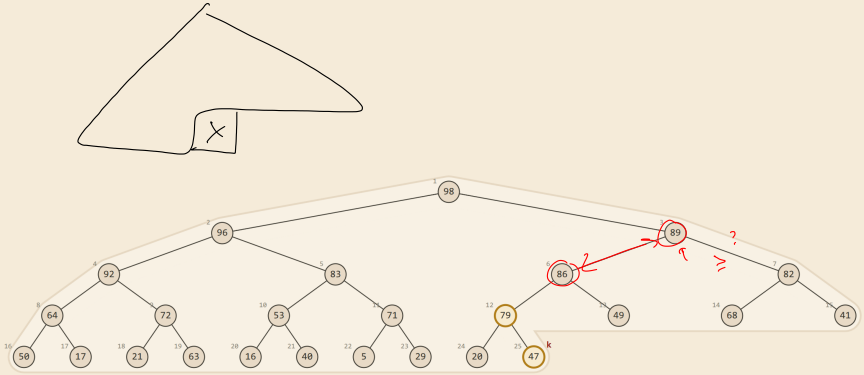


→ *sli.do/effalg*

# 3.4 Operations on Binary Heaps

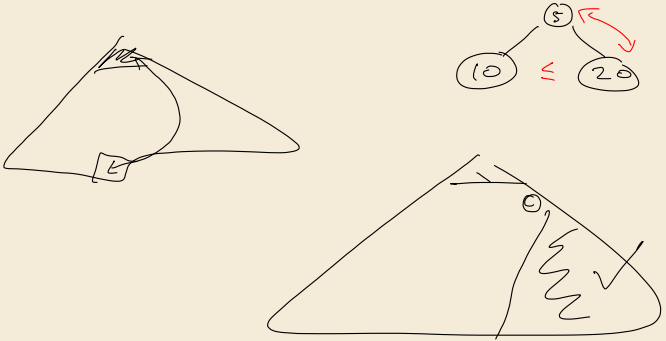
# Insert

1. Add new element at only possible place: bottom-most level, next free spot.
2. Let element *swim* up to repair heap order.



# Delete Max

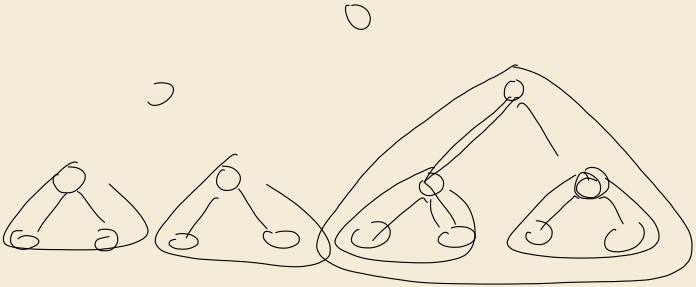
1. Remove max (must be in root).
2. Move last element (bottom-most, rightmost) into root.
3. Let root key *sink* in heap to repair heap order.



# Heap construction

- $n$ times `insert` $\rightsquigarrow$ *works, but we can do better!*
- instead:
  1. Start with singleton heaps (one element)
  2. Repeatedly merge two heaps of height $k$ with new element into heap of height $k+1$
     (= let the new element *sink*, exactly as in `delMax`)

  $\rightsquigarrow$ in the array: `sink(`$i$`)` for $i = \lfloor n/2 \rfloor, \ldots, 1$ (leaves need no sink)



# Analysis

**Height of binary heaps:**

- *height* of a tree: # edges on longest root-to-leaf path
- *depth/level* of a node: # edges from root $\rightsquigarrow$ root has depth $0$
- How many nodes on first $k$ *full* levels? $\sum_{\ell=0}^{k} 2^\ell = 2^{k+1} - 1$

$\rightsquigarrow$ Height of binary heap: $h = \min k$ s.t. $2^{k+1} - 1 \geq n$ $\Leftrightarrow$ $\lfloor \lg(n) \rfloor = h$

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**Analysis:**

- `insert`: new element "swims" up $\rightsquigarrow$ $\leq h$ steps ($h$ cmps)
- `delMax`: last element "sinks" down $\rightsquigarrow$ $\leq h$ steps ($2h$ cmps)
- `construct` from $n$ elements:

  cost = cost of letting *each node* in heap sink!

# Analysis

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  $$\leq 1 \cdot h + 2 \cdot (h-1) + 4 \cdot (h-2) + \cdots + 2^\ell \cdot (h-\ell) + \cdots + 2^{h-1} \cdot 1 + 2^h \cdot 0$$

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$$\leq 1 \cdot h + 2 \cdot (h-1) + 4 \cdot (h-2) + \cdots + 2^\ell \cdot (h-\ell) + \cdots + 2^{h-1} \cdot 1 + 2^h \cdot 0$$

$$= \sum_{\ell=0}^{h} 2^\ell (h-\ell) = \sum_{i=0}^{h} \frac{2^h}{2^i} i = 2^h \sum_{i=0}^{h} \frac{i}{2^i} \leq 2 \cdot 2^h \leq 2n \text{ steps } (\leq 4n \text{ comparisons})$$

# Binary heap summary

| Operation | Running Time |
|---|---|
| `construct(`$A[1..n]$`)` | $O(n)$ |
| `max()` | $O(1)$ |
| `insert(`$x$`,`$p$`)` | $O(\log n)$ |
| `delMax()` | $O(\log n)$ |
| `changeKey(`$x$`,`$p'$`)` | $O(\log n)$ |
| `isEmpty()` | $O(1)$ |
| `size()` | $O(1)$ |

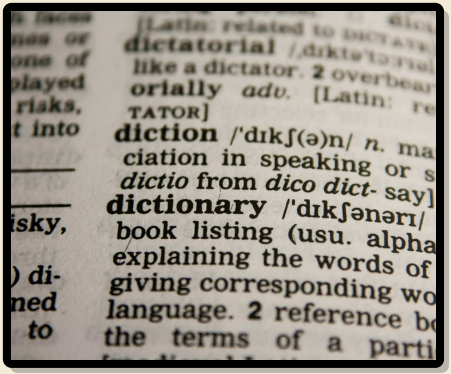
# 3.5 Symbol Tables

# Symbol table ADT

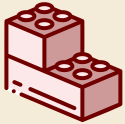
Java: `java.util.Map<K,V>`

**Symbol table / Dictionary / Map / Associative array / key-value store:**

Python dict `{k:v}`




- `put(`$k$`,`$v$`)` *Python* dict: `d[`$k$`] = `$v$
  Put key-value pair $(k, v)$ into table
- `get(`$k$`)` *Python* dict: `d[`$k$`]`
  Return value associated with key $k$
- `delete(`$k$`)` *Python* dict: `del d[`$k$`]`
  Remove key $k$ (any associated value) form table
- `contains(`$k$`)` *Python* dict: $k$ `in d`
  Returns whether the table has a value for key $k$
- `isEmpty(),size()`
- `create()`



*Most fundamental building block in computer science.*

(Every programming library has a symbol table implementation.)

# Symbol tables vs. mathematical functions

- similar interface
- but: mathematical functions are *static/immutable* (never change their mapping)
  (Different mapping is a *different* function)
- symbol table = *dynamic* mapping
  Function may change over time

# Elementary implementations

**Unordered (linked) list:**


Fast `put`


$\Theta(n)$ time for `get`

$\rightsquigarrow$ Too slow to be useful

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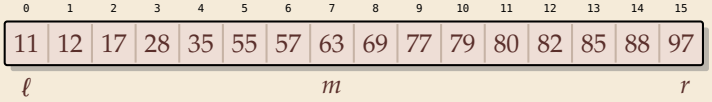
**Sorted *linked* list:**

- 👎 $\Theta(n)$ time for `put`
- 👎 $\Theta(n)$ time for `get`
- ⇝ Too slow to be useful
- ⇝ *Sorted order does not help us at all?!*

# Binary search

*It does help . . . if we have a sorted **array**!*

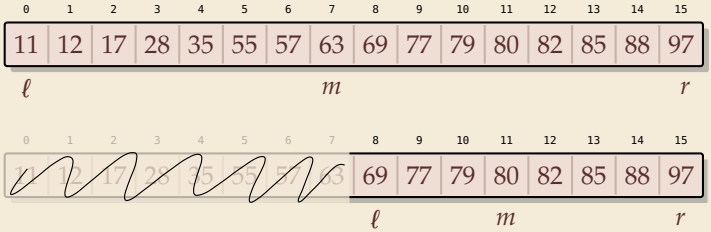
**Example:** search for 69



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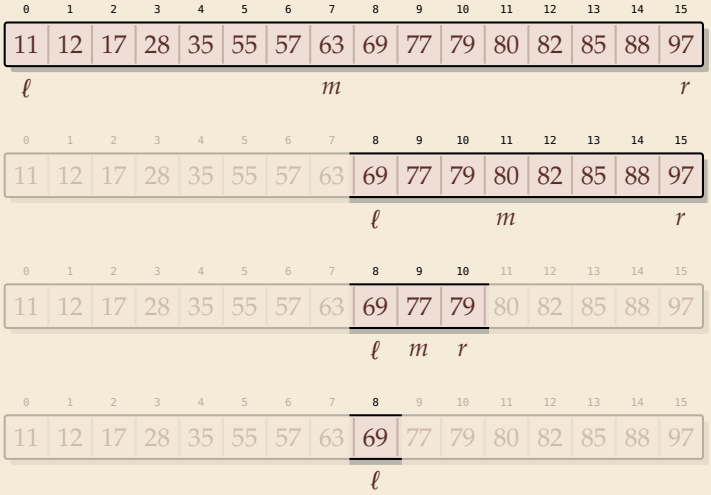
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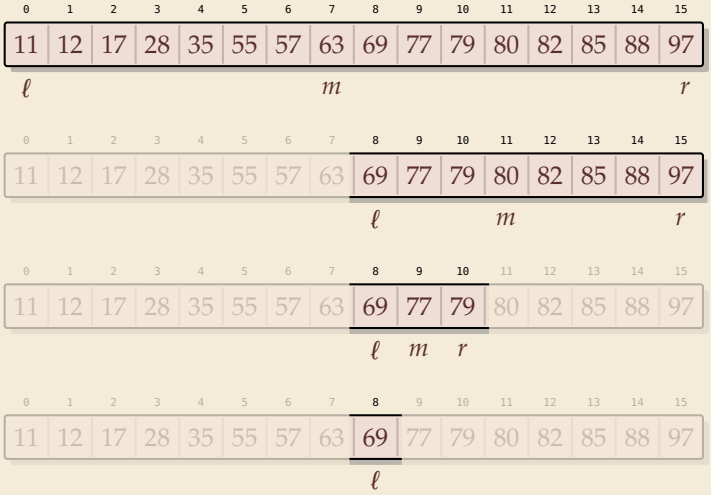
**Example:** search for 69



# Binary search

*It does help . . . if we have a sorted **array**!*

**Example:** search for 69




**Binary search**:

- halve remaining list in each step ($\pm 1$)

$\rightsquigarrow \leq \lfloor \lg n \rfloor + 1$ cmps in the worst case

needs random access!

# 3.6 Binary Search Trees

# Clicker Question



What is a binary search tree (tree in symmetric order)?

**A** A tree in which every node has exactly 2 children.

**B** A tree where all keys in the left subtree are smaller than the key at the root and all keys in the right subtree are bigger than the key at the root.

**C** A tree where all keys in the left subtree and right subtree are bigger than the key at the root.

**D** A tree that is stored in the heap-area of the memory.



→ *sli.do/effalg*

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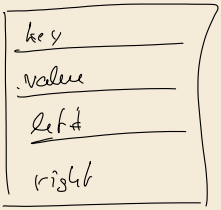
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→ *sli.do/effalg*

# Binary search trees

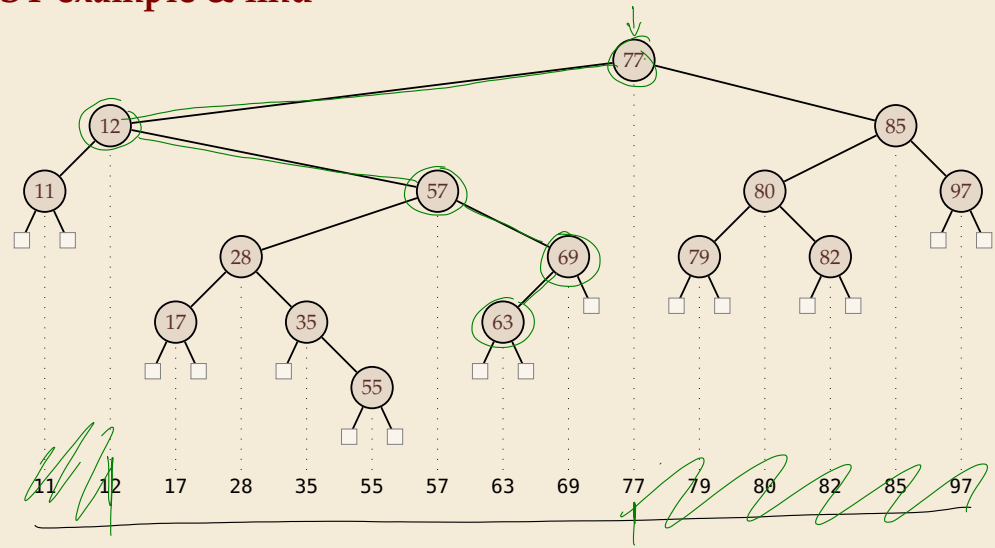
**Binary search trees (BSTs)** $\approx$ dynamic sorted array




- binary tree
  - Each node has left and right child
  - Either can be empty (`null`)
- Keys satisfy ***search-tree property***

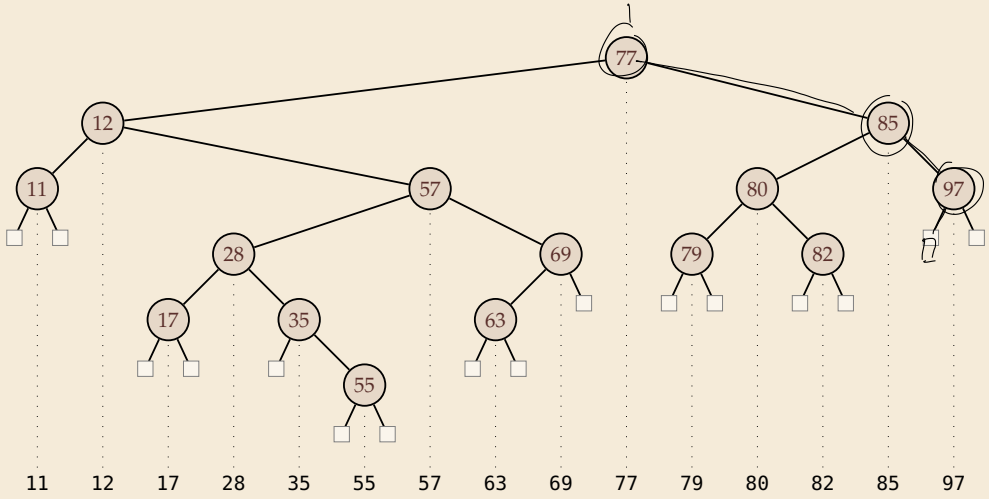
  all keys in left subtree $\leq$ root key $\leq$ all keys in right subtree

# BST example & find



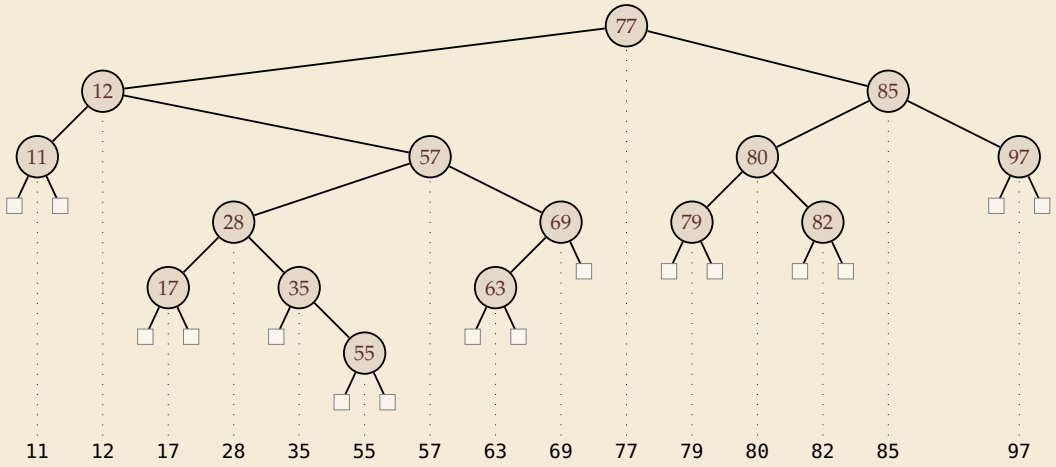
# BST insert

**Example:** Insert 88



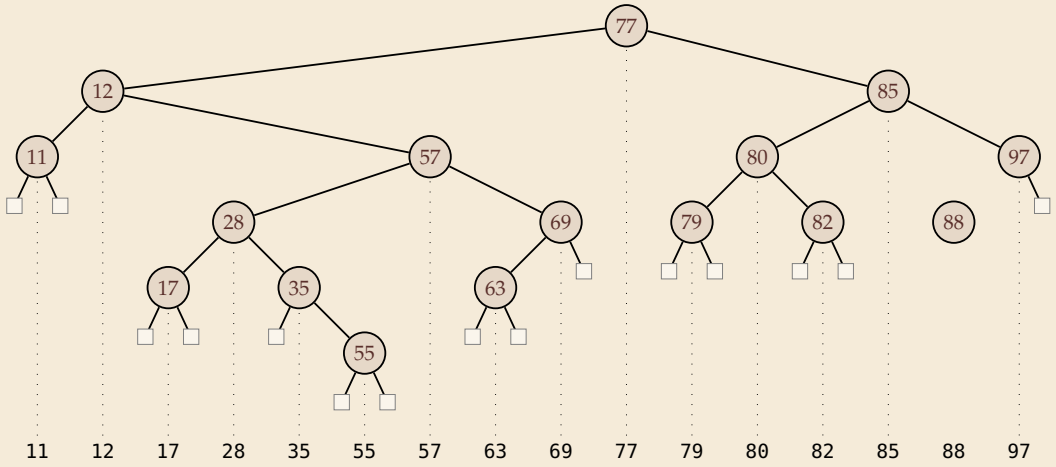
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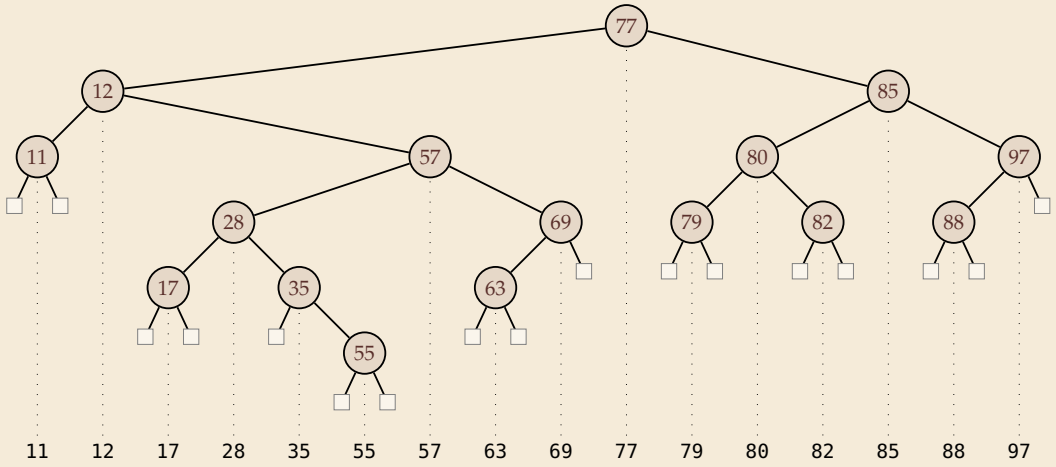
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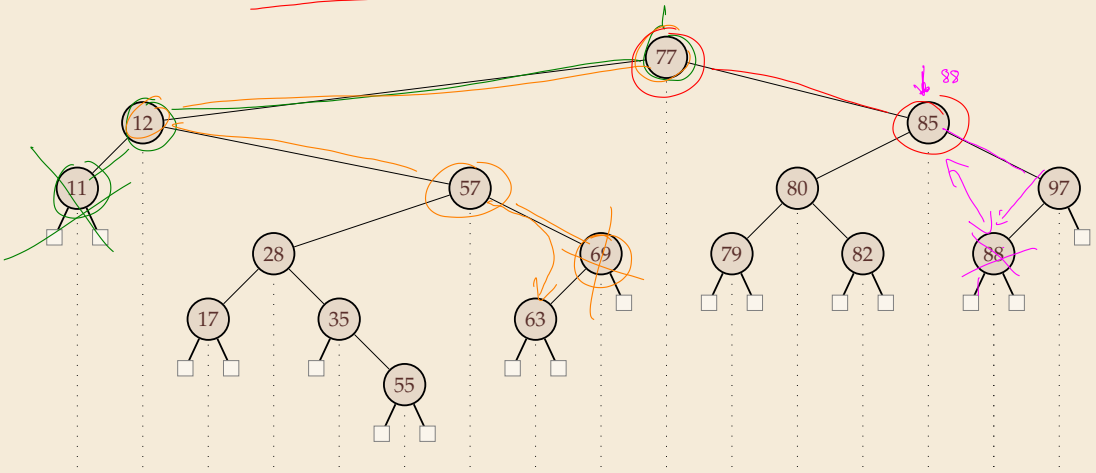
# BST insert

**Example:** Insert 88



# BST delete

- Easy case: remove leaf, e. g., 11 ⇝ replace by `null`
- Medium case: remove unary, e. g., 69 ⇝ replace by unique child
- Hard case: remove binary, e. g., 85 ⇝ swap with successor, recurse



# Analysis

Let $h$ be the *height* of the BST (longest root-to-leaf path, in edges).

- **Search:** follows one root-to-node path, one comparison per node

$\rightsquigarrow \leq h + 1$ comparisons $\rightsquigarrow$ $O(h)$

- **Insert:** unsuccessful search for key ($\leq h + 1$ cmps), then attach new leaf $\rightsquigarrow$ $O(h)$
- **Delete:**
  - search for the key ($O(h)$)
  - leaf / unary case: $O(1)$ pointer changes
  - binary case: find predecessor (walk down the left subtree, $O(h)$), swap, delete the predecessor (leaf or unary) $\rightsquigarrow$ $O(h)$

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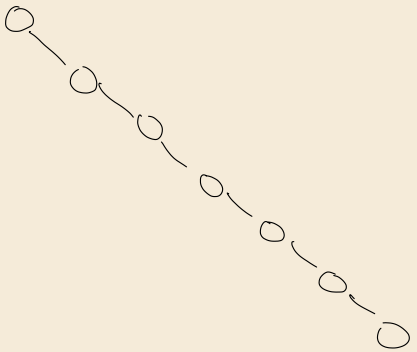
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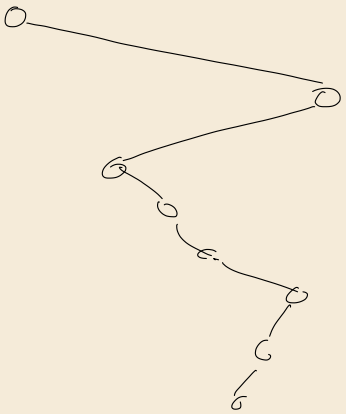
$\rightsquigarrow$ everything is $O(h)$ . . . *so how large is* $h$*?*

# What is the height of a BST?

**Worst Case:**

- $h = n - 1 = \Theta(n)$





Worst Case operations still $\Theta(n)$ !

balanced search trees (see below)

guess random is OK

# Random BSTs – Average Search Cost

**Model:** $n$ distinct keys inserted in *random order* (all $n!$ orders equally likely), no deletions.

Let $C_n$ := expected total cost of searching *each* key once (cost = #nodes on search path).

internal path length

equivalent to each key being chosen independent of all others

(iid from continuous)

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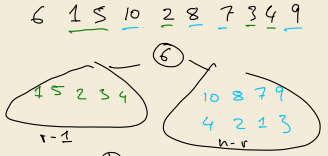
- root = first inserted key $\rightsquigarrow$ its rank $r$ is uniform in $[1..n]$
- left/right subtree: random BSTs of $r-1$ resp. $n-r$ keys
  (remaining keys still arrive in random order)

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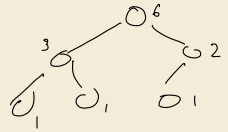
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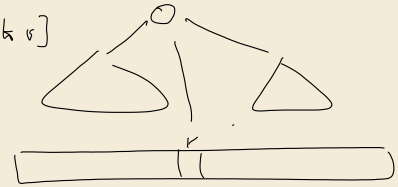



$$\rightsquigarrow\quad C_n \;=\; n \;+\; \frac{1}{n}\sum_{r=1}^{n}\big(C_{r-1}+C_{n-r}\big) \qquad (n \ge 1), \qquad C_0 = 0$$

comparison with root, once per key




$\frac{1}{n}$ = P[root has rank r]



# Random BSTs – Getting Rid of the Full History

$$C_n \;=\; n \;+\; \frac{1}{n}\sum_{r=1}^{n}\bigl(C_{r-1}+C_{n-r}\bigr) \;=\; n+\frac{2}{n}\sum_{k=0}^{n-1}C_k \qquad (n\geq 1), \qquad C_0 = 0$$

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$$\Bigg| \; + (n-1)C_{n-1}$$

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$$\frac{C_n}{n+1} \;=\; \frac{2n-1}{n(n+1)} \;+\; \frac{C_{n-1}}{n} \qquad (n\geq 1)$$

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$$\frac{C_n}{n+1} \;=\; \frac{2n-1}{n(n+1)} + \frac{C_{n-1}}{n} \qquad (n\geq 1)$$

Abbreviate $\boxed{D_n := \dfrac{C_n}{n+1}}$ $\qquad D_n = \dfrac{2n-1}{n(n+1)} + D_{n-1} \quad (n\geq 1), \qquad D_0 = 0$

$$D_n = \frac{2n-1}{n(n+1)} + \frac{2(n-1)-1}{(n-1)n} + \cdots + 0$$

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$$nC_n \;-\; (n-1)C_{n-1} \;=\; n^2-(n-1)^2 \;+\; \overbrace{2\sum_{k=0}^{n-1}C_k}^{=2C_{n-1}} \;-\; 2\sum_{k=0}^{n-2}C_k$$

$$\Longleftrightarrow \quad nC_n \;=\; 2n-1+(n+1)C_{n-1} \qquad \Big|\, : n(n+1)$$

$$\frac{C_n}{n+1} \;=\; \frac{2n-1}{n(n+1)} \;+\; \frac{C_{n-1}}{n} \qquad (n\geq 1)$$

Abbreviate $\boxed{D_n := \dfrac{C_n}{n+1}}$ $\qquad D_n = \dfrac{2n-1}{n(n+1)} + D_{n-1} \quad (n\geq 1), \qquad D_0 = 0$

**Note:** This is a telescoping recurrence!

# Random BSTs – Telescoping Recurrence

$$D_n = \sum_{k=1}^{n} \frac{2k-1}{k(k+1)}$$

# Random BSTs – Telescoping Recurrence

$$D_n = \sum_{k=1}^{n} \frac{2k-1}{k(k+1)}$$

$$\frac{2k-1}{k(k+1)} = \frac{-1}{k} + \frac{3}{k+1}$$ (partial fractions)

check: $\frac{-(k+1)+3k}{k(k+1)} = \frac{2k-1}{k(k+1)}$ ✓

$$\frac{2k-1}{k(k+1)} = \frac{a}{k} + \frac{b}{k+1} \quad \Big| \cdot k \Big|_{k=0}$$

$a = -1$

$$\frac{2k-1}{k(k+1)} = \frac{a(k+1)}{k} + \frac{b}{k+1} \quad \Big| (k+1) \Big|_{k=-1}$$

$b = 3$

# Random BSTs – Telescoping Recurrence

$$D_n = \sum_{k=1}^{n} \frac{2k-1}{k(k+1)} \qquad \frac{2k-1}{k(k+1)} = \frac{-1}{k} + \frac{3}{k+1}$$

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$$D_n = \underbrace{-\sum_{k=1}^{n} \frac{1}{k}}_{=H_n} + 3 \underbrace{\sum_{k=1}^{n} \frac{1}{k+1}}_{=H_{n+1}-1}$$

# Random BSTs – Telescoping Recurrence

$$D_n = \sum_{k=1}^{n} \frac{2k-1}{k(k+1)} \qquad \frac{2k-1}{k(k+1)} = \frac{-1}{k} + \frac{3}{k+1}$$

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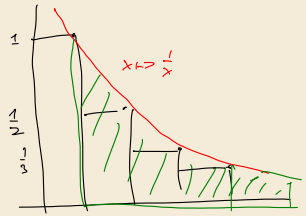
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$$C_n = (n+1)D_n = 2(n+1)H_n - 3(n+1) + 3 = 2(n+1)H_n - 3n$$

$$\boxed{H_n = \ln n + \gamma + O\left(\tfrac{1}{n}\right)}$$ (harmonic numbers; $\gamma \approx 0.5772$)

$$\rightsquigarrow \quad C_n \sim 2n \ln n \approx 1.39\, n \lg n$$ ($\ln n = \lg n / \lg e$, $2/\lg e \approx 1.386$)




$$H_n \approx \int_1^{n+1} \frac{1}{x}\, dx = \ln(n+1) - \ln(1)$$

# Random BSTs – Telescoping Recurrence

$$D_n = \sum_{k=1}^{n} \frac{2k-1}{k(k+1)} \qquad \frac{2k-1}{k(k+1)} = \frac{-1}{k} + \frac{3}{k+1}$$

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$\rightsquigarrow$ $C_n \sim 2n \ln n \approx 1.39\, n \lg n$ ($\ln n = \lg n / \lg e$, $2/\lg e \approx 1.386$)

$\rightsquigarrow$ average cost per search $\frac{C_n}{n} \sim 2 \ln n \approx 1.39 \lg n = \Theta(\log n)$

# Random BSTs – Telescoping Recurrence

$$D_n = \sum_{k=1}^{n} \frac{2k-1}{k(k+1)} \qquad \frac{2k-1}{k(k+1)} = \frac{-1}{k} + \frac{3}{k+1}$$ (partial fractions)

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(using $H_{n+1} = H_n + \frac{1}{n+1}$)

$$C_n = (n+1)D_n = 2(n+1)H_n - 3(n+1) + 3 = 2(n+1)H_n - 3n$$

$$\boxed{H_n = \ln n + \gamma + O\left(\tfrac{1}{n}\right)}$$ (harmonic numbers; $\gamma \approx 0.5772$)

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$\rightsquigarrow$ average cost per search $\frac{C_n}{n} \sim 2 \ln n \approx 1.39 \lg n = \Theta(\log n)$

$\neq$ *height*: $\mathbb{E}[h] \sim 4.311 \ln n$; also $\Theta(\log n)$, but much harder proof.

$\approx 29.85 \ln n$

Devroye: *A note on the height of binary search trees*, J. ACM 33(3) **1986**

# BST summary

| Operation | Running Time | Worst Case ($h = n - 1$) | Average Case (random insertion order) |
|---|---|---|---|
| `construct(`$A[1..n]$`)` | $O(nh)$ | $O(n^2)$ | $O(n \log n)$ |
| `put(`$k$`,`$v$`)` | $O(h)$ | $O(n)$ | $O(\log n)$ |
| `get(`$k$`)` | $O(h)$ | $O(n)$ | $O(\log n)$ |
| `delete(`$k$`)` | $O(h)$ | $O(n)$ | $O(\log n)$ |
| `contains(`$k$`)` | $O(h)$ | $O(n)$ | $O(\log n)$ |
| `isEmpty()` | $O(1)$ | $O(1)$ | $O(1)$ |
| `size()` | $O(1)$ | $O(1)$ | $O(1)$ |

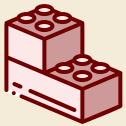
**Average case:**

- $n$ distinct keys inserted in random order, no deletions;
  then $\mathbb{E}[h] \sim 4.311 \ln n$ and the average search path has $\sim 2 \ln n \approx 1.39 \lg n$ nodes (see above).
- $h = \Theta(\log n)$ "with high probability":
  $\forall d \; \exists c : \Pr[h \geq c \lg(n)] \leq n^{-d}$
- With (Hibbard) deletions interleaved, the expected height degrades to $\Theta(\sqrt{n})$ – balanced BSTs (§3.8) fix both columns.

# 3.7 Ordered Symbol Tables

# Ordered symbol tables

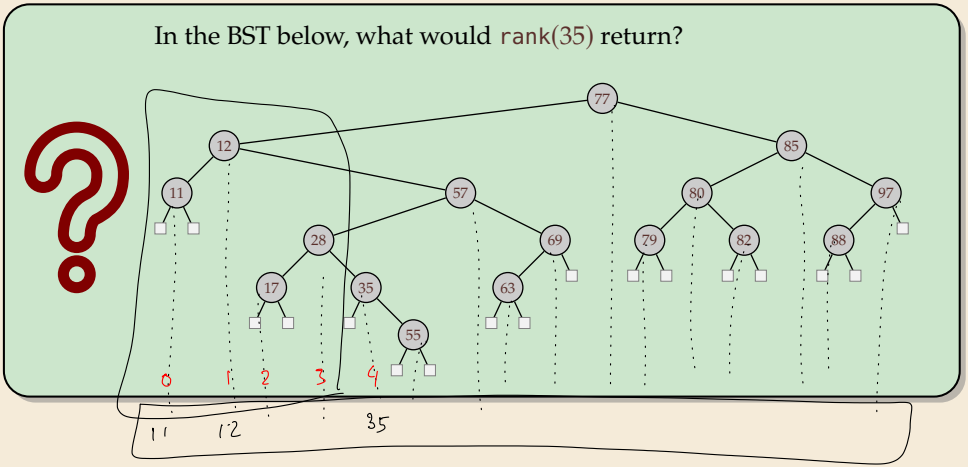
- `min()`, `max()`
  Return the smallest resp. largest key in the ST
- `floor(`$x$`)`, $\lfloor x \rfloor = \mathbb{Z}$.`floor(`$x$`)`
  Return largest key $k$ in ST with $k \leq x$.
- `ceiling(`$x$`)`
  Return smallest key $k$ in ST with $k \geq x$.
- `rank(`$x$`)`
  Return the number of keys $k$ in ST $k < x$.
- `select(`$i$`)`
  Return the $i$th smallest key in ST (zero-based, i. e., $i \in [0..n)$)



*With select, we can simulate access as in a truly dynamic array*!

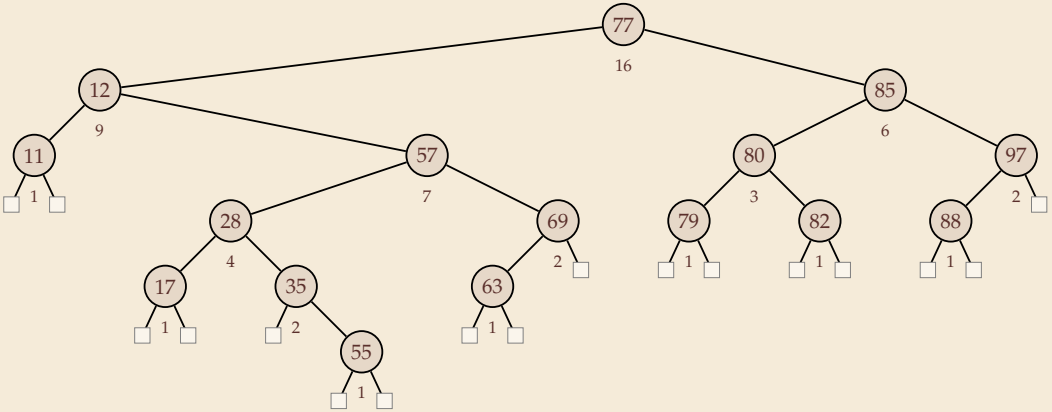
(Might not need any keys at all then!)

# Clicker Question

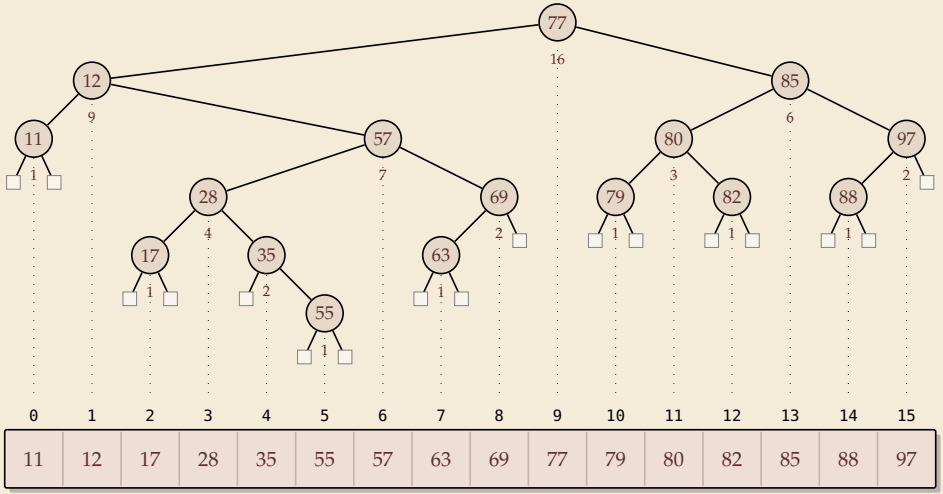



→ *sli.do/effalg*

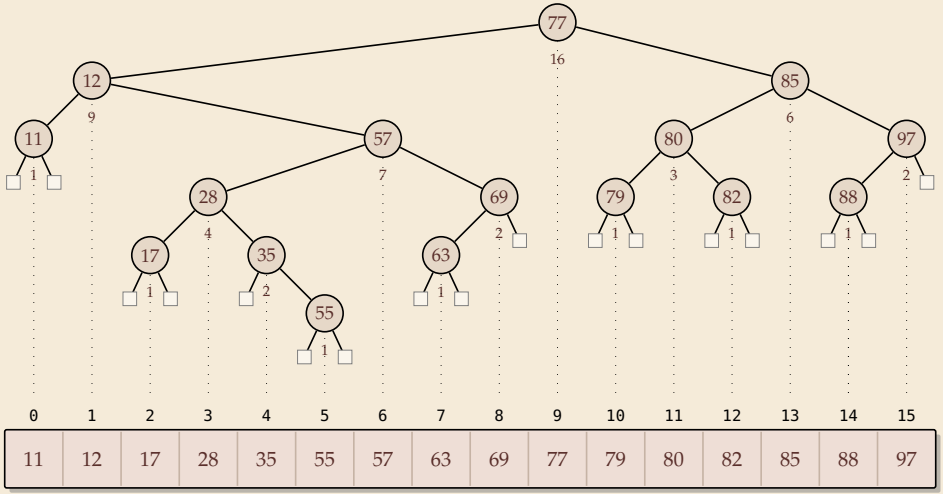
# Augmented BSTs



# Rank

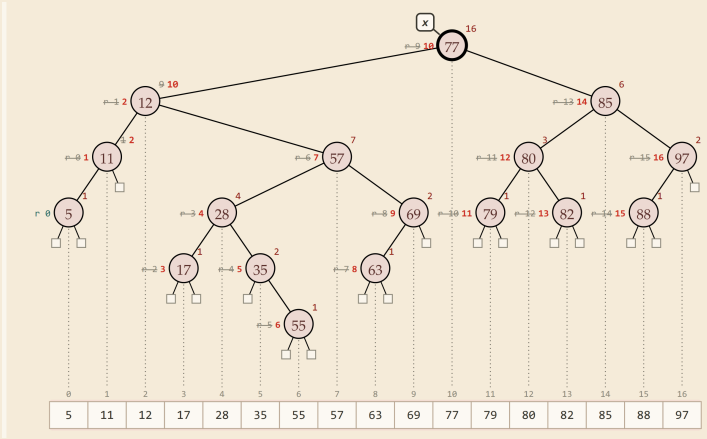

77
16
12
9
85
6
11
1
57
7
80
3
97
2
28
4
69
2
79
1
82
1
88
1
17
1
35
2
63
1
55
1
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
11 12 17 28 35 55 57 63 69 77 79 80 82 85 88 97

# Select


77
16
12
9
85
6
11
1
57
7
80
3
97
2
28
4
69
2
79
1
82
1
88
1
17
1
35
2
63
1
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1
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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# Why store subtree sizes?

- Note that in an augmented BST, each node stores the **size of its subtree**.
- . . . why not directly store the **rank**? Would make rank/select much simpler!



# Why store subtree sizes?

- Note that in an augmented BST, each node stores the **size of its subtree**.
- . . . why not directly store the **rank**? Would make rank/select much simpler!

- Problem: Single insertion/deletion can change *all* node ranks!

⇝ Cannot efficiently maintain node ranks.



Subtree sizes only change along search path ⇝ $O(h)$ nodes affected

# 3.8 Balanced BSTs

# Clicker Question



What ways of maintaining a **balanced** binary search tree do you know?
Write “none” if you have not seen balanced BSTs before.



→ *sli.do/effalg*

# Balanced BSTs

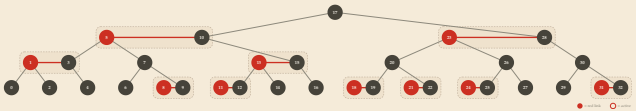
**Balanced binary search trees**:

- imposes shape invariant that guarantees $O(\log n)$ height
- adds rules to restore invariant after updates

# Balanced BSTs

**Balanced binary search trees**:

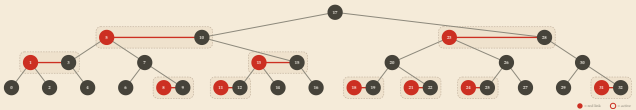
- imposes shape invariant that guarantees $O(\log n)$ height
- adds rules to restore invariant after updates
- many examples known
  - *AVL trees* (height-balanced trees)
  - *red-black trees*
  - *weight-balanced trees* (BB[$\alpha$] trees)
  - ...



# Balanced BSTs

**Balanced binary search trees**:

- imposes shape invariant that guarantees $O(\log n)$ height
- adds rules to restore invariant after updates
- many examples known
  - *AVL trees* (height-balanced trees)
  - *red-black trees*
  - *weight-balanced trees* (BB[$\alpha$] trees)
  - ...



**Other options:**

I'd love to talk more about some of these ...
⇝ *Advanced Data Structures*

- **amortization:** *splay trees*, *scapegoat trees*
  *COLA (cache oblivious lookahead array)*
- **randomization:** *randomized BSTs*, *treaps*, *skip lists*

# BSTs vs. Heaps

**Balanced binary search tree**

| Operation | Running Time |
|---|---|
| `construct(`$A[1..n]$`)` | $O(n \log n)$ |
| `put(`$k$`,`$v$`)` | $O(\log n)$ |
| `get(`$k$`)` | $O(\log n)$ |
| `delete(`$k$`)` | $O(\log n)$ |
| `contains(`$k$`)` | $O(\log n)$ |
| `isEmpty()` | $O(1)$ |
| `size()` | $O(1)$ |
| `min()` / `max()` | $O(\log n) \rightsquigarrow O(1)$ |
| `floor(`$x$`)` | $O(\log n)$ |
| `ceiling(`$x$`)` | $O(\log n)$ |
| `rank(`$x$`)` | $O(\log n)$ |
| `select(`$i$`)` | $O(\log n)$ |

**Binary heaps**

| Operation | Running Time |
|---|---|
| `construct(`$A[1..n]$`)` | $O(n)$ |
| `insert(`$x$`,`$p$`)` | $O(\log n)$ |
| `delMax()` | $O(\log n)$ |
| `changeKey(`$x$`,`$p'$`)` | $O(\log n)$ |
| `max()` | $O(1)$ |
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# BSTs vs. Heaps

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- apart from faster `construct`,
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- MaxPQ abstraction still helpful

# BSTs vs. Heaps

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~~**Binary heaps**~~ *Strict Fibonacci heaps*

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- apart from faster `construct`, BSTs always as good as binary heaps
- MaxPQ abstraction still helpful
- and faster heaps exist!

# 3.9 Hashing – Basics

# Lower bound for search

The fastest implementations of the ordered symbol table ADT require $\Theta(\log n)$ time to search among $n$ items. *Is this the best possible?*

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The fastest implementations of the ordered symbol table ADT require $\Theta(\log n)$ time to search among $n$ items. *Is this the best possible?*

- **Theorem:** In the comparison model (on the keys), $\Omega(\log n)$ comparisons are required to search a size-$n$ dictionary.

- *Proof:* Any algorithm defines a *decision tree*: internal nodes are comparisons, leaves are outcomes.
  There are $n + 1$ different outcomes ("is $x_i$" or "not found"), so $\geq n + 1$ leaves;
  a binary tree with $\ell$ leaves has height $\geq \lceil \lg \ell \rceil$ (ADS). ■

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  a binary tree with $\ell$ leaves has height $\geq \lceil \lg \ell \rceil$ (ADS). ■

$\rightsquigarrow$ Balanced BSTs are optimal . . . *in the comparison model.*

$\rightsquigarrow$ To beat $\Theta(\log n)$ we must leave that model:
*compute* with the key instead of only comparing it.

(and we will only get *expected* $O(1)$ – the lower bound is for the worst case)

# Hashing Recap

**Notation:** $U = [0..u)$ universe of keys; $x_1, \ldots, x_n$ the $n$ stored keys;
$T[0..m)$ the table; $\alpha = n/m$ the *load factor*;
$h : U \to [0..m)$

- store $x$ (and its value) at or near $T[h(x)]$

$\rightsquigarrow$ `get/put/delete` become array accesses ...

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$h : U \to [0..m)$

- store $x$ (and its value) at or near $T[h(x)]$

⇝ `get/put/delete` become array accesses ...

↯ ... if no two stored keys share an index.

# Hashing Recap

**Notation:** $U = [0..u)$ universe of keys; $x_1, \ldots, x_n$ the $n$ stored keys;
$T[0..m)$ the table; $\alpha = n/m$ the *load factor*;
$h : U \to [0..m)$

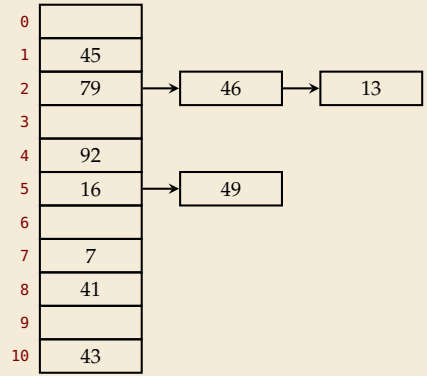
- store $x$ (and its value) at or near $T[h(x)]$

⇝ `get/put/delete` become array accesses . . .

↯ . . . if no two stored keys share an index.
$u \gg m$ ⇝ collisions *unavoidable* (pigeonhole),
and *likely*: already $n \approx 1.18\sqrt{m}$ random keys
collide with probability $\geq \frac{1}{2}$ (birthday paradox)

- Collision handling (details: ADS / S&W §3.4)
  - *separate chaining*: $T[i]$ holds list keys with $h(x) = i$
  - *open addressing*: probe $T[h(x)], T[h(x)+1], \ldots$
    linear, or other probe sequences
    (quadratic, Robin Hood, cuckoo, . . . )

$m = 11$, $h(x) = x \bmod 11$; chaining



# Hashing Recap [2]

**Notation:** $U = [0..u)$ universe of keys; $x_1, \ldots, x_n$ the $n$ stored keys;
$T[0..m)$ the table; $\alpha = n/m$ the *load factor*;
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- *rehash* into a new table when $\alpha$ leaves $[\alpha_{\min}, \alpha_{\max}]$
  $\rightsquigarrow$ $\Theta(n + m)$, amortized $O(1)$ (doubling trick!)

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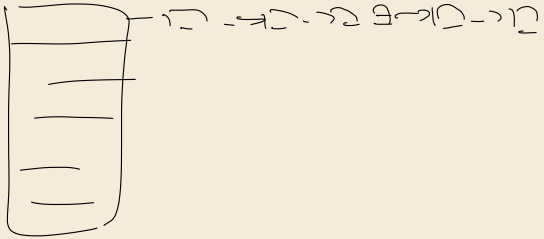
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- **Cost** of an operation $\approx$ #keys examined $=$ length of the chain / probe sequence.

  *How long are they?*

  - in worst case: all keys(!)



# Uniform hashing assumption

- ***Simple Uniform Hashing Assumption:***
  For the keys $x_1, \ldots, x_n$ in our dictionary we have

(unrealistic (but still bound.)

$$\mathop{\mathbb{P}}_{x_i \neq x_j} \left[ h(x_i) = h(x_j) \right] = \frac{1}{m}$$

$\approx$ keys in the input are drawn i.i.d. uniformly from $U = [0..u)$, and $h$ spreads $U$ evenly over $[0..m)$.

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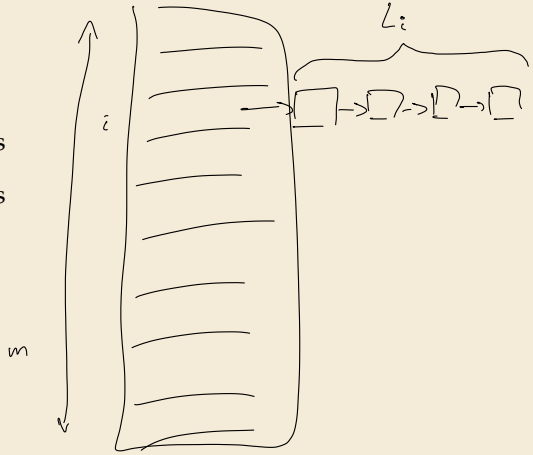
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and $h$ spreads $U$ evenly over $[0..m)$.

- This is an assumption about the *input*. We will come back to that.

# Analysis of chaining (average case)

- $L_i$: # keys in bucket $i$
  - how many keys hash to $i$?
- unsuccessful search: $1 + L_{h(x)}$ memory accesses
- successful search: $\leq 1 + L_{h(x_i)}$ memory accesses

$\rightsquigarrow$ key quantity: $\mathbb{E}[L_{h(x_i)}]$



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$$= \begin{cases} 1 & h(x_i) = h(x_j) \\ 0 & \text{else} \end{cases}$$

- Define $X_{i,j} = \mathbb{1}\{h(x_i) = h(x_j)\} \rightsquigarrow L_{h(x_i)} = \sum_{j=1}^{n} X_{i,j}$
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&\underset{(*)}{=} 1 + \sum_{j \neq i} \frac{1}{m} = 1 + \frac{n-1}{m} \\
&\leq 1 + \alpha
\end{aligned}
$$

# Clicker Question



What can we say about the **distribution** of $L_i$?
Select all correct statements.

- **A** $L_0, \ldots, L_{m-1}$ have the same (marginal) distribution.
- **B** $L_0, \ldots, L_{m-1}$ are (stochastically) independent.
- **C** $L_{h(x_i)}$ is $\mathrm{Bin}(n, \frac{1}{m})$ distributed
- **D** $L_{h(x_i)} - 1$ is $\mathrm{Bin}(n-1, \frac{1}{m})$ distributed

Binomial distribution

$\mathrm{Bin}(n,p)$ = # heads in $n$ coin flips w/ biased coin where heads has prob. $p$.



→ *sli.do/effalg*

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# 3.10 Universal Hashing

# Discussion

- Running time for hashing with chaining is $O(1)$ if we keep $m$ proportional to $n$

⇝ see amortized analysis of the *doubling trick*

- `java.util.HashMap` works like this!

+ red-black tree for large buckets

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- **But:** only in expectation, and only under the **assumption** of *uniform hashing*!

  - Worst case: all keys collide in one bucket.
  - For *every fixed* $h$ this worst case exists: $u \geq n \cdot m \quad \rightsquigarrow$ some bucket contains $\geq n$ keys of $U$ (pigeonhole)
  - An adversary who knows $h$ inserts exactly those $\rightsquigarrow$ every operation $\Theta(n)$.
    (Not hypothetical: hash-flooding attacks on Perl, PHP, Java `hashCode`, ... – cf. Sedgewick & Wayne)
  - *Can we avoid this?*

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*Randomization!*



# Universal families of hash functions

- **Idea:** Choose $h$ **at random** from a set $\mathcal{H}$.
- A set $\mathcal{H} \subseteq \{h : U \to [0..m)\}$ of hash functions is ***c-universal*** if for a (uniformly) random $h \in \mathcal{H}$ we have

$$\forall x \neq x' \ : \ \mathop{\mathbb{P}}_{h \in \mathcal{H}}\left[h(x) = h(x')\right] \ \leq \ \frac{c}{m}$$

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$$\mathbb{P}_{x_i \neq x_j}\big[h(x_i) = h(x_j)\big] = \frac{1}{m}$$

$\rightsquigarrow$ essentially the same collision probability!

**But:** instead of *hoping* for a benign input (average case), we *enforce* the average case by randomization – for *every* input.

# Analysis of chaining (randomized)

- Choose $h$ *at random* from a $c$-universal $\mathcal{H}$.
  *Then* let the adversary pick the keys $x_1, \ldots, x_n$ and $x$. w/o knowing h.
- $L_i$: # keys in bucket $i$
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&\underset{\text{lin. } \mathbb{E}}{=} \sum_{j=1}^{n} \mathbb{E}[X_{i,j}] \\
&= \sum_{j=1}^{n} \mathbb{P}[X_{i,j} = 1] \\
&= 1 + \sum_{j \neq i} \mathbb{P}[h(x_i) = h(x_j)] \\
&\underset{(*)}{\leq} 1 + \sum_{j \neq i} \frac{c}{m} \\
&\leq 1 + c\alpha
\end{aligned}
$$

Same analysis – only *what is random* has changed.

Adversary may know $\mathcal{H}$, but not which $h$ we drew; re-draw $h$ at every rehash.

# Do universal families exist?

- Yes! e. g., the set of **all** functions $\mathcal{H} = \{h : U \to [0..m)\}$
  - With $h \in \mathcal{H}$ uniform, hash values behave exactly like i.i.d. uniform random variables: $\forall x \in U : h(x) \overset{\mathcal{D}}{=} \text{Uniform}[0..m)$

  $\rightsquigarrow \mathop{\mathbb{P}}_{h \in \mathcal{H}}\left[h(x) = h(x')\right] = \frac{1}{m}$

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- too expensive! We need $\Theta(u)$ space to store $h$ ...

$|\mathcal{H}| = m^u$ ↯

$\log_2 \uparrow = u \cdot \log_2 m$ bits

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- Wanted: a *small* family $\rightsquigarrow$ $h$ fits in $O(1)$ words and is $O(1)$ to evaluate

# Dot-product hashing

- **Assumption:** $m$ is prime (Bertrand's postulate: $[n, 2n)$ contains a prime)
- **Assumption:** $u = m^r$ for some $r \in \mathbb{N}_{\geq 1}$

$\rightsquigarrow$ key $x$ corresponds to a number with $r$ digits in base $m$: $x = \langle x_{r-1}, x_{r-2}, \ldots, x_0 \rangle$

- For $a \in [0..u)$, define $h_a(x) = (a^T x) \bmod m = \sum_{l=0}^{r-1} a_l x_l \bmod m$

  $(a^T x)$: dot product of $a$ and $x$ as $r$-dimensional vectors

- $\mathcal{H} = \{h_a \mid a \in [0..u)\}$

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- $\mathcal{H} = \left\{ h_a \,\middle|\, a \in [0..u) \right\}$

- **Properties:**
  - $h_a$ can be stored in $O(\log u)$ bits (store $a$!)
  - $h_a$ is computed in $O(1)$ time in the word-RAM with word size $w = \Omega(\log u)$

    $\rightsquigarrow$ keys fit in $O(1)$ words

# Dot-product hashing is universal

**Theorem 3.1**

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# Linear congruential hashing

## Theorem 3.2 (Linear congruential hashing)

Let $p \geq u$ be prime and $m \leq p$. Then

$$\mathcal{H}_{p,m} = \left\{ h_{a,b} \;\middle|\; a \in [1..p),\ b \in [0..p),\ \ h_{a,b}(x) = \big((ax + b) \bmod p\big) \bmod m \right\}$$

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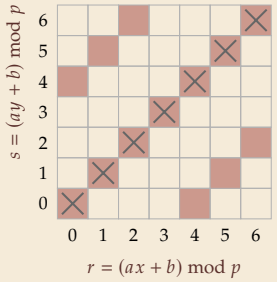
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- Further, $(a,b) \mapsto (r,s)$ is a *bijection* onto the $p(p-1)$ pairs $r \neq s$: $a = (r-s)\,(x-y)^{-1} \bmod p, \quad b = (r - ax) \bmod p$.

all pairs $(r,s)$ for $p = 7$, $m = 4$




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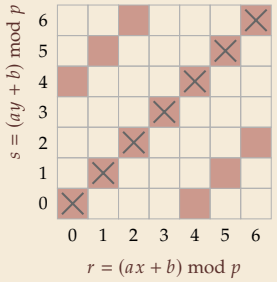
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$\rightsquigarrow$ uniform $(a,b)$ = uniform *off-diagonal* cell $(r,s)$.

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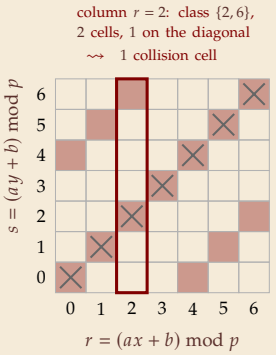

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**Proof (cont.):**

- collision iff $r \equiv s \pmod{m}$: the shaded cells.

  Per column $r$: a residue class of $\leq \lceil p/m \rceil$ cells,
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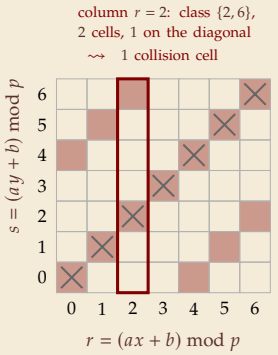
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  shaded off-diagonal: $2+2+2+0 = 6$ of 42 cells $\rightsquigarrow \quad \frac{1}{7} \leq \frac{1}{4}$ ✓




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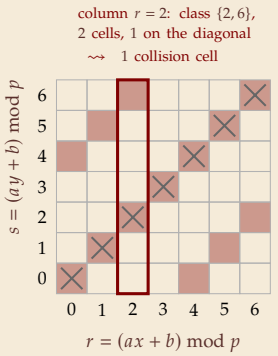
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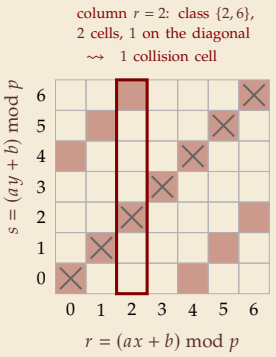
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**Note:** *Crude count* (all shaded cells, diagonal included, $a = 0$ allowed): 13 of $49 \approx 0.27 > \frac{1}{4}$ $\rightsquigarrow$ only $c$-universal with $c = \left(\lceil \frac{p}{m} \rceil / \frac{p}{m}\right)^2 \le 4$. The bijection is what lets us drop the diagonal.

# Other universal families

## Theorem 3.3 (Multiplicative hashing)

For $w$-bit keys and $m = 2^d$, the family
$\left\{ h_z(x) = \left((z \cdot x) \bmod 2^w\right) \operatorname{div} 2^{w-d} \;\middle|\; z \text{ odd},\ z \in [1..2^w) \right\}$ is 2-universal. ◀



**Dietzfelbinger, Hagerup, Katajainen, Penttonen**: *A Reliable Randomized Algorithm for the Closest-Pair Problem*, J. Algorithms **1997**

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*In practice:* Python randomizes its string hash per process (SipHash, since 3.3) for exactly this reason; Java does not, and instead turns long chains into red-black trees.

# Hashing with chaining – Summary

**Average case → expectation in the worst case**

- With a *universal hash family* we achieve the same complexity as under the uniform hashing assumption –

  **but:** without any assumption on the input!

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- total running time is still amortized (doubling trick for rehashing!)


fastest (amortized) access time


simple data structure


somewhat wasteful with space


worst case $O(n)$ per operation (cuckoo hashing fixes this for searches)

# Comparison of Classic Hashing Schemes

| Hash table design | Search hit | Search miss | Insert | Space | good $\alpha = \frac{n}{m}$ |
|---|---|---|---|---|---|
| Separate Chaining | $\approx 1 + \frac{1}{2}\alpha$ | $\approx \alpha$ | = miss | $n + m$ | $\approx 2$ |
| Linear Probing | $\approx \frac{1}{2}\left(1 + \frac{1}{1-\alpha}\right)$ | $\approx \frac{1}{2}\left(1 + \frac{1}{(1-\alpha)^2}\right)$ | = miss | $m$ | $\leq 0.5$ |
| Quadratic Probing | $\approx 1 + \ln\left(\frac{1}{1-\alpha}\right) - \frac{1}{2}\alpha$ | $\approx \frac{1}{1-\alpha} - \alpha + \ln\left(\frac{1}{1-\alpha}\right)$ | = miss | $m$ | $\leq 0.7$ |
| Robin Hood (Lin. P.) | = Lin. P.; Var. $O(1)$ | $\leq$ Lin. P. | = miss | $m$ | $\leq 0.9$ |
| Robin Hood Clever | $O(1)$ | $O(1)$ | = miss | $m$ | $\leq 1$ (=any!) |
| $d$-way Cuckoo H. | $\leq d$ worst case | $\leq d$ worst case | amort. | $m$ | $< c_d$ |

$c_2 = \frac{1}{2}$, $c_3 \approx 0.918$

- Assumption: uniform hashing (all $m^n$ hash sequences equally likely)
- Cost: expected # (equality) comparisons, for $n, m \to \infty$ with $\alpha$ fixed
- Space usage in words on top of space for items (without space for optional optimizations)
- Robin Hood only *reorders* within clusters: same mean as linear probing, but the cost of a search is *concentrated* (and with random probing + "mean-centred" search even $O(1)$ expected in a full table [Poblete & Viola 2019])

# Comparison of Classic Hashing Schemes

| Hash table design | Search hit | Search miss | Insert | Space | good $\alpha = \frac{n}{m}$ |
|---|---|---|---|---|---|
| Separate Chaining | $\approx 1 + \frac{1}{2}\alpha$ | $\approx \alpha$ | = miss | $n + m$ | $\approx 2$ |
| Linear Probing | $\approx \frac{1}{2}\left(1 + \frac{1}{1-\alpha}\right)$ | $\approx \frac{1}{2}\left(1 + \frac{1}{(1-\alpha)^2}\right)$ | = miss | $m$ | $\leq 0.5$ |
| Quadratic Probing | $\approx 1 + \ln\left(\frac{1}{1-\alpha}\right) - \frac{1}{2}\alpha$ | $\approx \frac{1}{1-\alpha} - \alpha + \ln\left(\frac{1}{1-\alpha}\right)$ | = miss | $m$ | $\leq 0.7$ |
| Robin Hood (Lin. P.) | = Lin. P.; Var. $O(1)$ | $\leq$ Lin. P. | = miss | $m$ | $\leq 0.9$ |
| Robin Hood Clever | $O(1)$ | $O(1)$ | = miss | $m$ | $\leq 1$ (=any!) |
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More improvements possible with word-RAM bitwise tricks $\rightsquigarrow$ *Advanced Data Structures*

# Hashing vs. Balanced Search Trees

**Advantages of Balanced Search Trees**

- $O(\log n)$ worst-case operation cost
- Does not require any assumptions, special functions, or known properties of input distribution
- Predictable (and often smaller) space usage (exactly $n$ nodes)
- Never need to rebuild the entire structure
- supports ordered dictionary operations (rank, select etc.)

**Advantages of Hash Tables**

- $O(1)$ operations (if hashes well-spread and load factor small)
- We can choose space-time tradeoff via load factor
- Cuckoo hashing achieves $O(1)$ worst-case for search & delete